Kwan Man (Nick) Cheng

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Vision-Language Learning | Agentic Systems | Post-Training | Robot Learning

EDUCATION

Carnegie Mellon University — **Aug 2026 - May 2028**
M.S. in Robotics — Pittsburgh, PA

University of Wisconsin-Madison — **Jan 2024 - May 2026**
B.S. in Computer Science (transferred from Beijing Normal University), GPA: 3.98/4.00 — Madison, WI
Teaching Assistant, Matrix Methods in Machine Learning | Dean's List (all semesters)

University of California, Berkeley — **Jan 2023 - Jan 2024**
Exchange & Summer Program, GPA: 3.93/4.00 — Berkeley, CA
A+ grades in Data Structures and Introduction to Artificial Intelligence (Top 1%)

RESEARCH & PROFESSIONAL EXPERIENCE

Carnegie Mellon University - Robotics Institute — **Apr 2026 - Present**
Research Assistant | Prof. Fernando De la Torre — Pittsburgh, PA

- Developing an agentic framework using coding agents and LLM reflection to evolve reusable robot skills integrated with VLA/WAM models, improving OOD task success from 0% to 72.0% in LIBERO-Recover.
- Built SkillWeaver, a VLM-guided search and PPO skill pipeline, generating 31K demonstrations across 11K+ scenes (8.7M frames), raising average LIBERO-PRO-Object OOD success from 47.6% to 78.1%, and demonstrating successful sim-to-real transfer on a Franka Panda.
- Generated 128K spatially grounded reasoning commands from 7K+ synthetic trajectories to train hierarchical VLM-VLA policies, improving LIBERO-PRO success from below 10% to 70.8%.

University of Wisconsin-Madison - Video Lab — **Jan 2025 - May 2026**
Research Assistant | Prof. Yin Li — Madison, WI

- Co-first-authored DRIFT, a residual flow-matching adapter that refines coarse VLM predictions into precise continuous outputs for robot control and temporal/spatial grounding.
- Trained DRIFT with Qwen3-VL-2B and OpenVLA. Qwen3-VL-2B + DRIFT achieved 97.9% average success on LIBERO and 61.5% on SIMPLER WidowX.
- Built video preprocessing and LoRA fine-tuning pipelines using 20K videos and 100K temporal annotations.

Qualcomm — **Sep 2025 - Dec 2025**
Machine Learning Engineer (Capstone Project) — Madison, WI

- Built a multimodal presentation coach with Qwen3-VL, distilled speech-feedback training data, and exported Qwen3-VL-2B to ONNX for Snapdragon NPU deployment.

Tsinghua University — **May 2024 - May 2025**
Research Assistant | Prof. Wenwu Zhu and Prof. Bing Xu — Beijing, China (Hybrid)

- Co-developed Modular-Cam, an LLM-guided video diffusion framework using temporal transformers and LoRA modules to control camera movements across multiple scenes.
- Designed a two-stage U-Net for nighttime-light reconstruction with multi-scale fusion and cross-resolution attention, reducing county-level RMSE by 48.6% versus the strongest baseline.

Quwan Technology — **Nov 2024 - Jan 2025**
Algorithm Intern — Guangzhou, China

- Built multi-node inference infrastructure for text-to-video generation with PyTorch Distributed.
- Developed training data pipelines and fine-tuned text-to-video models on proprietary datasets.

SELECTED PUBLICATIONS

*DRIFT: A Residual Flow Adapter for Decoding Continuous Outputs in Vision-Language Models**. Under review at **ICLR 2027.**
SkillWeaver: Agentic Exploration over Neural Interaction Skills for Scalable Robot Data Generation. **CoRL 2026.**
Modular-Cam: Modular Dynamic Camera-view Video Generation with LLM. **AAAI 2025.**
*An Extended VIIRS-like Artificial Nighttime Light Data Reconstruction (1986-2024)**. **Scientific Data, 2026.**
** Co-first author.*

OPEN SOURCE PROJECTS

Honing Infra (101 stars). Infrastructure for agent optimization. Implemented GEPA for harness evolution. Evolved a Red Alert 2 agent and its harness through expert-guided SFT and GEPA, raising win rate from 0% to 100% against the baseline opponent.

TECHNICAL SKILLS

Machine Learning: PyTorch, LLMs/VLMs/VLAs, multimodal learning, diffusion models, flow matching, LoRA, reinforcement learning, distributed training, transformers, PPO/DPO/GRPO, DAgger, SFT, GEPA
Robotics & Simulation: Isaac Lab, Isaac Sim, LIBERO, SIMPLER, ROS, Blender, inverse kinematics
Languages & Systems: Python, Java, JavaScript, C/C++, Docker, ONNX, Unity, AWS, GCP